

Analyzing Medical Insurance Costs

Smoking Emerges as the Strongest Predictor of Elevated Insurance Costs

July 2025

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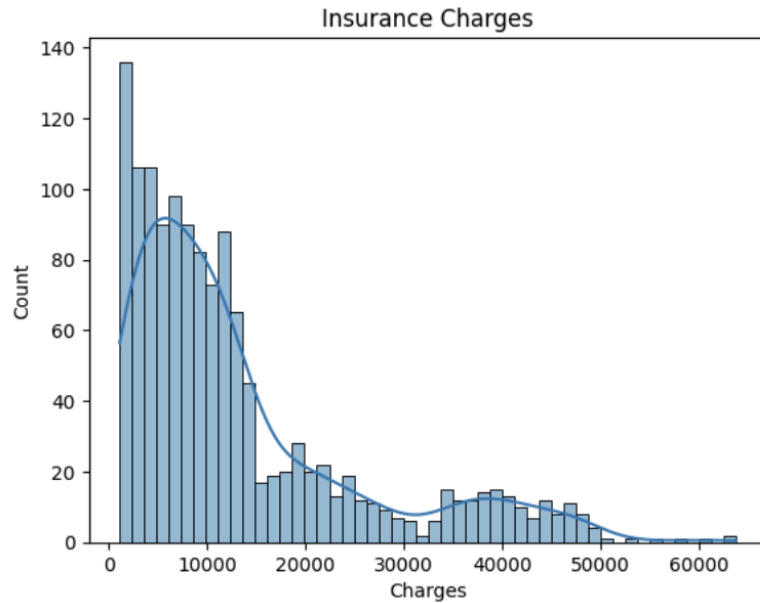
Introduction

This project explores the relationships between individual health and lifestyle factors and medical insurance costs, using the Medical Cost Personal Datasets. Our primary aim was to uncover patterns and assess predictive variables that influence insurance charges, especially those related to smoking habits, body mass index (BMI), age, and geographic region. By taking a hypothesis-driven approach rooted in exploratory data analysis (EDA), we were able to generate targeted insights that could enhance pricing strategies, inform healthcare policy, and drive risk-aware decision-making within the insurance sector.

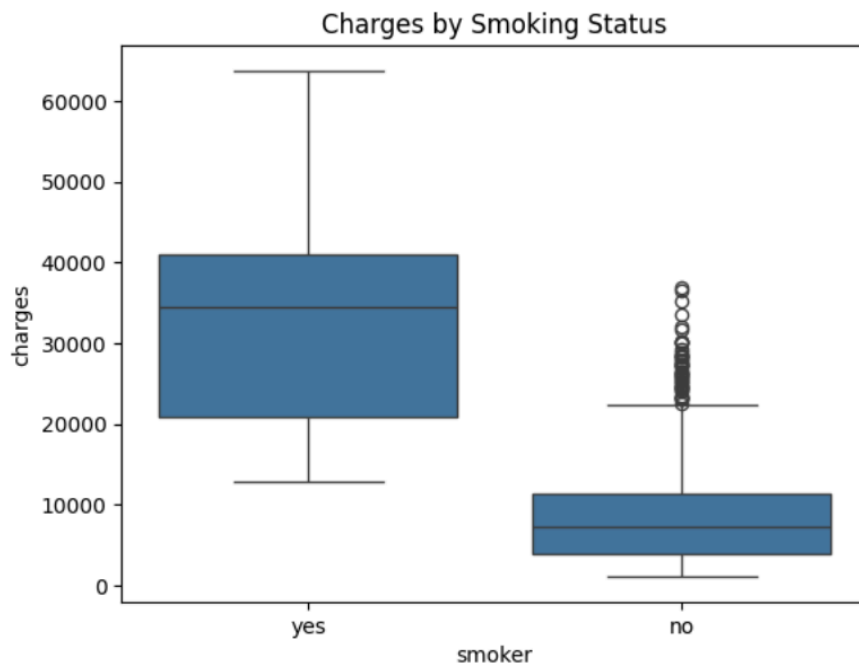
Research Process

We began our research process by assessing the distribution and range of key features. Most individuals had BMI values clustered between the mid-20s and low-30s, which suggests a central tendency around the overweight category. However, multiple entries exceeded BMI values of 40 or 50, which were flagged as clear outliers indicative of obesity-related health risks. These extreme values not only widened the charge distribution but also helped focus our analysis on the connection between obesity and medical costs.

Our analysis of insurance charges revealed that most fall below \$20,000, with the largest concentration below \$10,000. Beyond \$20,000, fewer individuals are represented, yet some reach nearly \$60,000 in premiums, reinforcing a right-skewed distribution. A KDE (Kernel Density Estimation) curve visualized this shape, confirming that a small group bears significantly higher costs while most remain within a lower bracket.



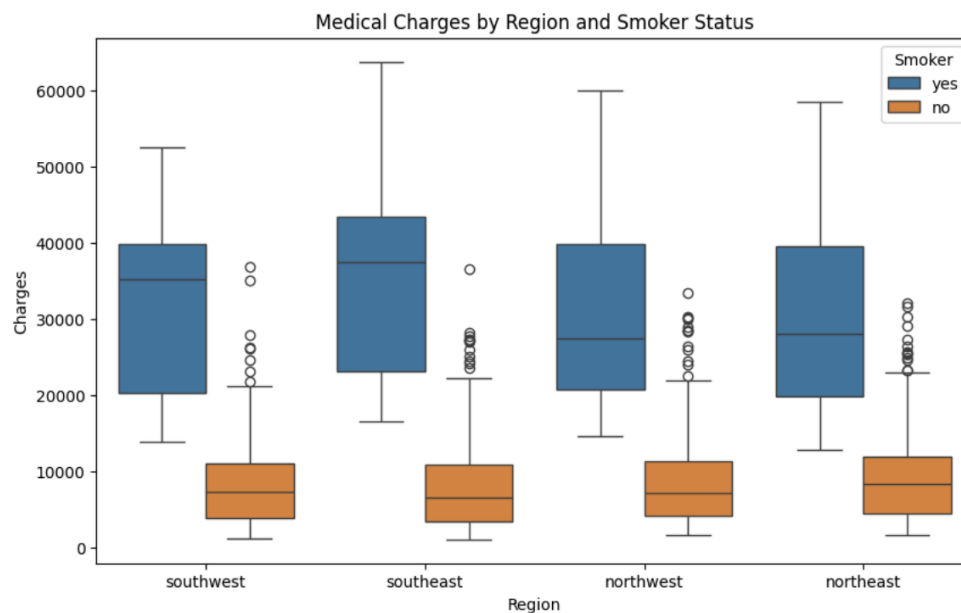
We also evaluated categorical features to check for demographic balance. Notably, the dataset is roughly evenly split between male and female individuals, suggesting no gender bias, which allows for more objective modeling. Smoking status, however, was far from balanced, roughly 75–80% of individuals are non-smokers, and only 20–25% are smokers. This disparity proved meaningful: smoking status showed strong correlation with insurance charges, and its uneven distribution provided a powerful predictor variable.



Dataset Description & Initial Exploration

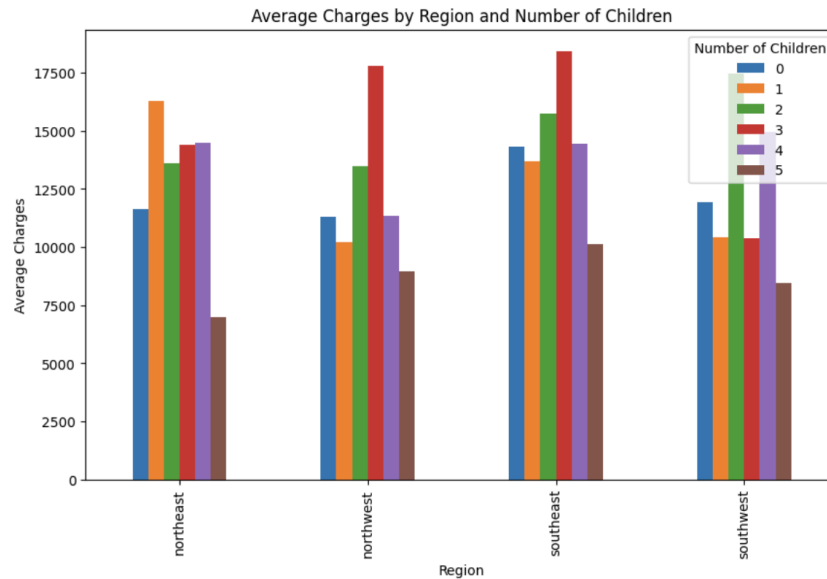
Our hypothesis surrounding obesity was validated through charge comparisons across BMI categories. Individuals classified as obese consistently paid higher median charges than those in normal or overweight groups. The obese category also presented a broader spread and more extreme outliers, some approaching \$60,000. In contrast, normal and overweight groups had lower medians and narrower interquartile ranges, reflecting more predictable costs. Underweight individuals represented a small portion of the dataset and displayed modest charges.

Smoking was one of the strongest indicators of higher costs. Across the dataset, smokers incurred significantly higher charges than non-smokers, regardless of region. This pattern held especially true in the Southeast, which not only had the highest average medical charges but also the sharpest contrast between smokers and non-smokers. The top five most expensive individuals in the dataset were all smokers, reinforcing this trend.



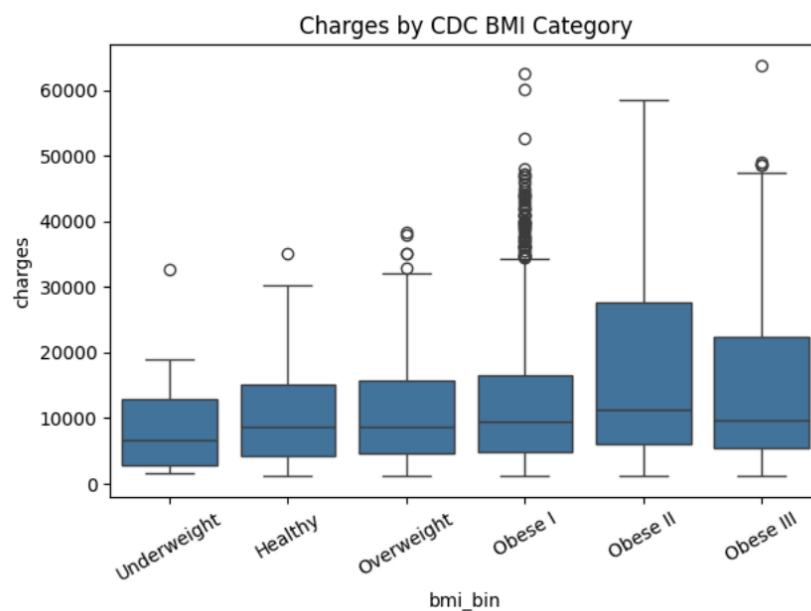
Age demonstrated a moderate positive correlation with insurance costs: older individuals tend to pay more, although this link was not as strong as BMI or smoking. Interestingly, within the most expensive group, age was broadly distributed, suggesting that smoking and BMI are more dominant cost drivers.

When exploring the impact of family dynamics, the number of children showed no consistent increase in charges. Individuals with five children surprisingly incurred the lowest costs. However, those with two or four children did exhibit higher charges than those with zero, one, three, or five, suggesting a non-linear and possibly lifestyle-related interaction.

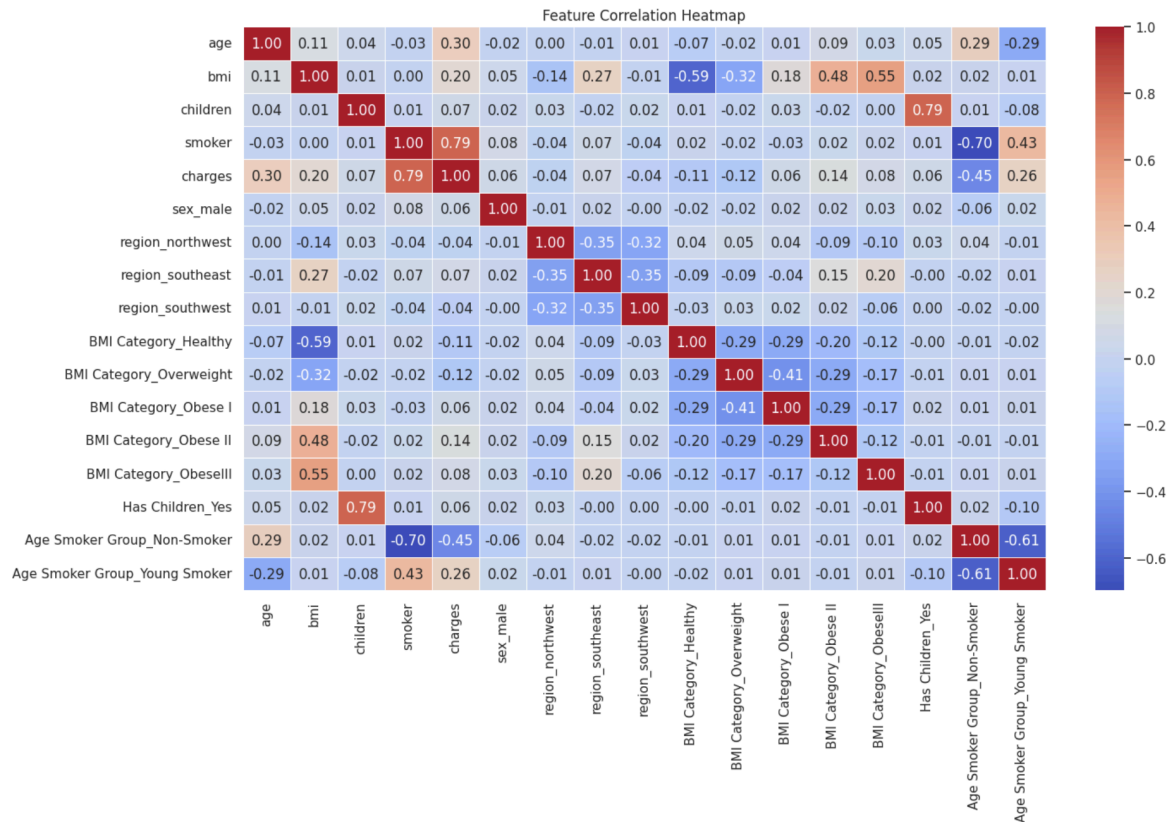


Findings & Feature Impact Analysis

Our analysis revealed smoking status as a dominant factor affecting insurance costs. Smokers showed significantly higher mean and median charges, which persisted across age groups. BMI demonstrated a moderate positive correlation with charges, especially among individuals classified as obese.



In exploring feature interactions, we found compounding effects, for instance, smokers with high BMI paid substantially more than any other group. This guided our recommendation for targeted health interventions in high-risk populations.

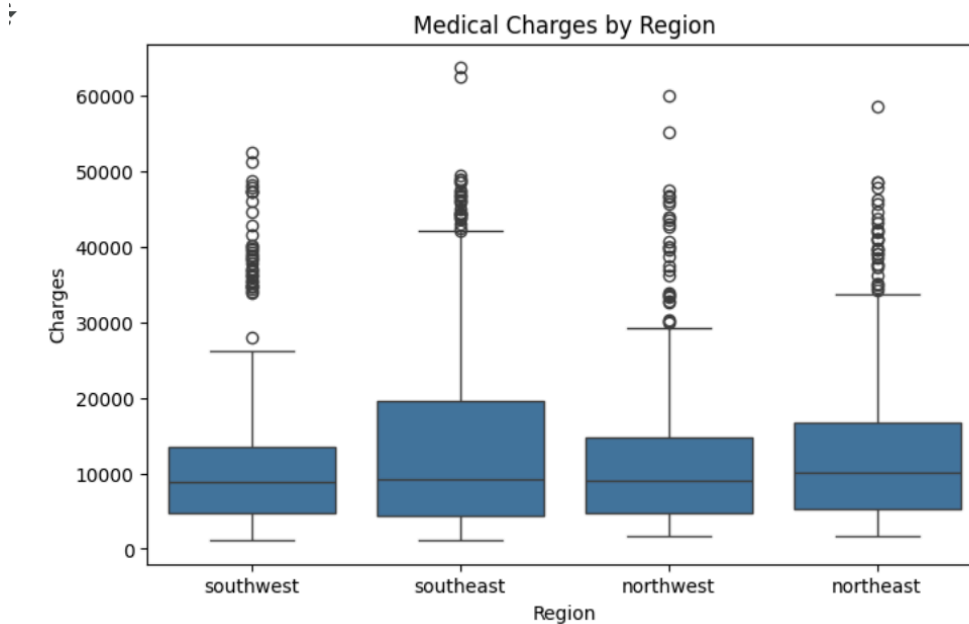


Regional Insights & Trend Analysis

Region-based segmentation yielded insightful trends. The Southeast had the highest count of entries and consistently reported the highest average and median charges. This region also displayed the most skewed distribution, most individuals incurred low costs, but several extreme outliers lifted the average substantially.

In contrast, the Southwest demonstrated the narrowest range and most consistent charge values, with fewer outliers and more tightly packed data. This stability makes it potentially easier to model and predict costs for that region.

Both the Northwest and Northeast exhibited moderate representation and charge ranges, with the Northeast showing the highest median values among all four regions. This could suggest that while outliers affect the Southeast's averages, the Northeast may be facing broader underlying healthcare cost pressures.



This finding suggests that insurers may benefit from location-aware pricing models, and that public health efforts should consider local contexts when targeting cost reduction strategies.

Conclusion

Through systematic analysis and feature exploration, we identified smoking status, BMI, and geographic region as the most impactful variables affecting medical insurance charges. Our process, grounded in careful distribution analysis, visualization, and statistical correlation, uncovered trends that not only confirm well-known healthcare patterns but also reveal nuanced interactions. These insights highlight opportunities for insurers to develop more targeted pricing models, while also pointing to public health efforts that could reduce costs through smoking cessation and obesity intervention programs. Further research could incorporate socioeconomic data or expand toward predictive modeling using regression or classification algorithms.